SVD

$$\boxed{X_{N\times p} = U_{N\times p} \Sigma_{p\times p} V^T_{p\times p}}$$

$UU^T = I, \quad VV^T = I$

$\Sigma = \text{dia}(\sigma_1, \sigma_2, \cdots, \sigma_r)$

$r = \text{rank}(X), \quad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r$

$U = [u_1, u_2 \cdots u_p] \qquad u_i \in \mathbb{R}^N$

$V = \begin{bmatrix} v_1^T \\ v_2^T \\ \vdots \\ v_p^T \end{bmatrix} \qquad v_j \in \mathbb{R}^p$

$X = \sum_{i=1}^{r} \sigma_i u_i v_i^T$ Rank-1 decomposition

$X_k = \sum_{i=1}^{k} \sigma_i u_i v_i \qquad X_k \rightsquigarrow X$ as $k \rightsquigarrow r$

Eg $X = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$ rank$(X) = 2$

$A = X^T X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$

$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$

det $X = 3$

Trace $X = 4$

$Ae_1 = \begin{pmatrix} 2 \\ 1 \end{pmatrix} \quad Ae_2 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$

$Ae_1 + Ae_2 = \begin{pmatrix} 3 \\ 3 \end{pmatrix}$

$\Rightarrow A(e_1 + e_2) = 3\begin{pmatrix} 1 \\ 1 \end{pmatrix}$

$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \lambda_1 = 3$

$\boxed{\lambda_2 = 1}$ $\qquad v_2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \lambda_2 = 1$

$\sigma_1 = \sqrt{\lambda_1} = \sqrt{3}, \quad \sigma_2 = \sqrt{\lambda_2} = 1$

$v_1 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ 1/\sqrt{2} \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{pmatrix}$

$Xv_1 = (U \Sigma V^T) v_1$

$= [u_1 \; u_2] \begin{bmatrix} \sigma_1 & 0 \\ 0 & \sigma_2 \end{bmatrix} \begin{bmatrix} v_1^T \\ v_2^T \end{bmatrix} v_1$

$= [u_1, u_2] \begin{bmatrix} \sigma_1 v_1^T \\ \sigma_2 v_2^T \end{bmatrix} v_1 = \sigma_1 u_1$

$$u_1 = \frac{1}{\sigma_1} X v_1 \qquad \langle u_1, u_1 \rangle = \frac{1}{\sigma_1^2} \langle X v_1 \; X v_1 \rangle$$

$$= \frac{1}{\sigma_1^2} v_1^T \underbrace{X^T X v_1}$$

$$= \frac{1}{\sigma_1^2} (v_1^T \sigma_1^2 v_1) = 1.$$

$||^y \quad u_2 = \frac{1}{\sigma_2} X v_2$

$$u_1 = \cancel{x} \frac{1}{\sqrt{3}} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix} = \frac{1}{\sqrt{6}} \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$u_2 = \frac{1}{\cancel{\sigma_2}} \; \frac{1}{1} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ -1 \\ 1 \end{bmatrix}$$

$$\underline{u_3 = u_1 \times u_2} = \cdots \frac{1}{\sqrt{3}} \begin{bmatrix} 1 \\ -1 \\ -1 \end{bmatrix}$$

$$U = [u_1 \;\; u_2 \;\; u_3] \qquad \Sigma = \begin{bmatrix} \sqrt{3} & \\ & 1 \end{bmatrix} \quad V = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}$$

<u>Check</u> $\boxed{X = U \Sigma V^T}$ <u>Verify</u>

<u>Recall</u> Multi-linear regression

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_n x_n + \varepsilon$$

$$y = X\beta + \varepsilon$$

$$\varepsilon = \| y - X\beta \|^2$$

$$\boxed{X\beta = y} \qquad \underset{n \times p}{X} : \mathbb{R}^n \to \mathbb{R}^p \qquad \boxed{\hat{\beta} = (X^T X)^{-1} X^T \beta}$$

lin. Least square <u>soln</u>

<u>Generalized or pseudoinverse of X</u>

It turns out that $\boxed{\hat{\beta} = X^{\dagger} y}$

generalized inverse of X

$X = U\Sigma V^T$

$X^+ = (U\Sigma V^T)^+ = (V^T)^+ \Sigma^+ U^+$

$= V \Sigma^+ U^T$

$$\Sigma^+ = \left[\begin{array}{cccc|c} 1/\sigma_1 & & & & \\ & 1/\sigma_2 & & & 0 \\ & & \ddots & & \\ & & & 1/\sigma_r & \\ \hline & & 0 & & 0 \end{array}\right]$$

Ex Find the generalized inverse of $\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$

Ex Finding least square soln of (in python)

$2x_1 - x_2 + x_3 = 1$

$x_1 + x_2 - x_3 = 7$

$3x_1 + x_2 + x_3 = 10$

$4x_1 - x_2 - x_3 = 0$

$5x_1 + 2x_2 + 3x_3 = -5$

$\boxed{X\beta = y}$

$X = \begin{bmatrix} 2 & -1 & 1 \\ \vdots & & \vdots \\ 5 & 2 & 3 \end{bmatrix}$ $y = \begin{bmatrix} 1 \\ 7 \\ 10 \\ 0 \\ -5 \end{bmatrix}$

$\beta = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$

$$\boxed{\hat{\beta} = (X^TX)^{-1} X^T y \equiv X^+ y}$$

PCA $X_{N\times p} \rightsquigarrow$ centred around origin (normalized)

$C_X := \frac{1}{N-1} X^T X$

covariance matrix of X.

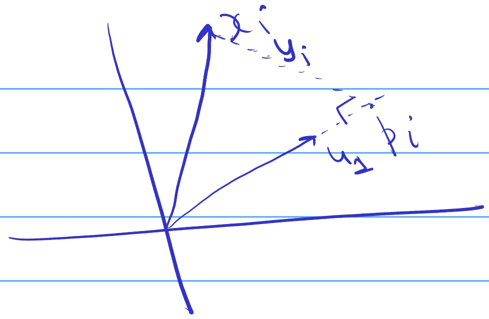
eigen-decomposition of C_X

$C_X = Q \Lambda Q^T$

columns of Q are principal components

$Q = [q_1\ q_2 \cdots q_p]_{p\times p}$ $C_X q_i = \lambda_i q_i$

$\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p$.



$p_i = \underline{(x_i^T u_1)}\, u_1$

$\|y_i\|^2 = \|x_i - u_1\|^2$

$$\min \sum_{i=1}^{n} \|x_i - u_1\|^2 \quad \text{s.t.} \quad \|u_1\| = 1$$

$$\min \sum (x_i - u_1)^T (x_i - u_1)$$

$$= \min \sum x_i^T x_i - x_i^T u_1 + u_1^T x_1 + \underbrace{u_1^T u_1}_{1}$$

$$= \min (-2 \sum x_i^T u_1)$$

$$= \max \sum x_i^T u_1$$

Relationship between SVD & PCA

$$C_x = \frac{1}{N-1} X^T X = Q \Lambda Q^T$$

on the other hand

$$X = U \Sigma V^T \quad (\text{SVD of } X)$$

$$C_x = \frac{1}{N-1} (U\Sigma V^T)^T (U \Sigma V^T)$$

$$= \frac{1}{N-1} V \Sigma^T \Sigma V^T$$

$$\Rightarrow \quad Q \Lambda Q^T = \frac{1}{N-1} V \Sigma^T \Sigma V^T = \frac{1}{N-1} V \Sigma^2 V^T$$

$$\boxed{\Lambda = \frac{1}{N-1} \Sigma^2}$$

$$\boxed{V = Q}$$

Transformed data

matrix of principal components = V

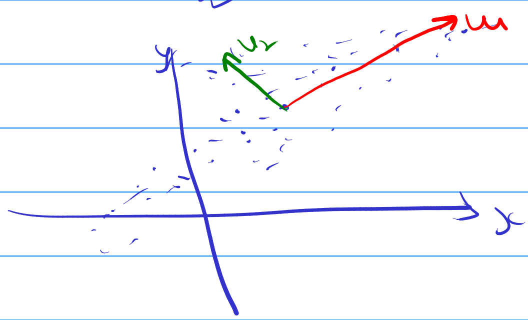
$$x_i = V z_i \qquad 1 \le i \le n.$$

$$X^T = V Z^T$$

$$\Rightarrow \quad V^T X^T = V^T (V Z^T) = Z^T$$

$$\Rightarrow \boxed{XV = Z}$$

or $$XV = (U \Sigma V^T) V = \boxed{U\Sigma = Z}$$



Kernel PCA

$X = \begin{bmatrix} x_1^T \\ \vdots \\ x_n^T \end{bmatrix}$ $x_i \in \mathbb{R}^p$

$\Phi \quad X_{N\times p} \longrightarrow \mathcal{H}$

$K_{ij} = K(x_i, x_j) := \underline{\langle \phi(x_i), \phi(x_j) \rangle}$

Kernels

Linear: $K(x,y) = x^T y$

Poly $K(x,y) = (x^T y + \alpha)^d$

RBF = $K(x,y) = e^{-\gamma \|x-y\|^2}$

$\underline{X - \mu} \qquad x_i - \mu$

$\mu_\phi = \frac{1}{N} \sum_{i=1}^{N} \phi(x_i)$

$\tilde{\Phi}(x_i) = \phi(x_i) - \mu_\phi$

$$\tilde{K}_{ij} = \langle \tilde{\varphi}(x_i), \tilde{\varphi}(x_j) \rangle$$

$$= \langle \phi(x_i) - \mu_\phi, \varphi(x_j) - \mu_\phi \rangle$$

$$= \langle \underline{\varphi(x_i)} - \frac{1}{N} \sum_k \varphi(x_k), \varphi(x_j) - \frac{1}{N} \sum_{k=1}^{N} \phi(x_k) \rangle$$

$$= \langle \varphi(x_i), \varphi(x_j) \rangle - \frac{1}{N} \sum_{k=1}^{N} \langle \varphi(x_i), \varphi(x_k) \rangle$$

$$- \frac{1}{N} \sum_{k=1}^{N} \langle \varphi(x_k), \varphi(x_j) \rangle + \frac{1}{N^2} \sum_l \sum_k \langle \varphi(x_k), \varphi(x_l) \rangle$$

$$\tilde{K}_{ij} = K_{ij} - \frac{1}{N} \sum_k K_{ik} - \frac{1}{N} \sum K_{kj} + \frac{1}{N^2} \sum_l \sum_k K_{kl}$$

$$\boxed{\tilde{K} = K - \mathbb{1}K - K\mathbb{1} + \mathbb{1}K\mathbb{1}}$$

$\mathbb{1} = \frac{1}{N} e e^T \qquad e = \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix}_{N\times 1}$

$e = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

$ee^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}_{3\times 3}$